

BST Vol 8 Ch 9 — Continuum Mechanics + Elasticity (v0.3.1, Calibration #23 substance refill)

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Vol 8 Chapter 9 — Continuum Mechanics + Elasticity

Headline result

Continuum mechanics treats matter as continuous distributions of mass + momentum, ignoring atomic discreteness. The key tensors are:

Stress tensor σ_{ij} : force per unit area on a surface with normal in j -direction, component in i -direction. Symmetric ($\sigma_{ij} = \sigma_{ji}$) for non-couple-stress materials.

Strain tensor $\epsilon_{ij} = (1/2)(\partial u_i / \partial x_j + \partial u_j / \partial x_i)$: symmetric measure of deformation.

Hooke's law (linear elasticity): $\sigma_{ij} = C_{ijkl} \epsilon_{kl}$ where C_{ijkl} is the 4th-rank elastic stiffness tensor (21 independent components for triclinic; fewer for higher symmetry).

For isotropic materials, only 2 independent elastic constants (e.g., Young's modulus E + Poisson ratio ν , or Lamé constants $\lambda + \mu$).

BST framework: continuum mechanics inherits from substrate-fluid framework (Vol 6 Stat Mech, Lyra) + substrate-K-type structure. Honest scope: full substrate-derivation of elastic constants for specific materials requires multi-week per Vol 8 weak-foundation scope.

Substrate mechanism

Stress + strain tensors as substrate-natural:

In substrate framework, deformation is described by displacement field $u(x)$ on substrate-coordinate manifold. Strain tensor ϵ_{ij} quantifies local deformation; stress tensor σ_{ij} encodes internal forces. Hooke's law relates them linearly for small deformations.

Elastic constants:

For isotropic materials: - Young's modulus E (stress/strain in 1D tension) - Poisson ratio ν (transverse contraction in tension) - Shear modulus $G = E/(2(1+\nu))$ - Bulk modulus $K = E/(3(1-2\nu))$

Relations: $K = \lambda + (2/3)\mu$; $G = \mu$; $\nu = \lambda/(2(\lambda+\mu))$; $E = \mu(3\lambda+2\mu)/(\lambda+\mu)$.

Substrate-K-type framework: - Elastic moduli connect to substrate energy-density framework - BST primary integer ratios appear in selected materials (multi-week per Vol 9 Ch 3 cross-link) - Phonon coupling (Vol 9 Ch 7) $\rightarrow$ elastic moduli via lattice dynamics

Wave propagation in elastic media

Longitudinal waves (compression): velocity $c_L = \sqrt{(K + 4G/3)/\rho} = \sqrt{((\lambda + 2\mu)/\rho)}$.

Transverse waves (shear): velocity $c_T = \sqrt{G/\rho} = \sqrt{(\mu/\rho)}$.

In solids, $c_L > c_T$ (compression faster than shear). In liquids, $c_T = 0$ (no shear support).

Match precision

I-tier framework. Standard continuum mechanics + linear elasticity preserved at any precision (engineering applications). Substrate-derivation of specific material elastic constants multi-week.

Cross-volume dependencies

- Vol 6 (Stat Mech, Lyra): substrate fluid framework + thermodynamic potentials
- Vol 8 Ch 7 (Rigid Body Mechanics): rigid body = elastic body in $K \rightarrow \infty$ limit
- Vol 9 Ch 3 (Electron Band Structure): lattice $\rightarrow$ elastic moduli via phonon dynamics
- Vol 9 Ch 6 (Magnetism): magnetoelastic coupling
- Vol 9 Ch 7 (Phonons): acoustic phonons = long-wavelength elastic waves

K-audit anchor

K237 Vol 8 Ch 9 Continuum + Elasticity K-audit pre-stage (Keeper pending).

Pedagogical walkthrough (3-level)

Level 1 — Bright 5th-grader

When you stretch a rubber band, it pushes back. When you press on a sponge, it squishes. Continuum mechanics describes how materials deform under forces — the “stress” (force per area) and “strain” (how much it’s deformed). For small deformations, Hooke’s law says they’re proportional ($\sigma = E \varepsilon$). BST acknowledges this framework comes from substrate’s fluid description; full substrate-derivation of specific material elastic constants is honest multi-week work.

Level 2 — Undergraduate physics student

Stress tensor σ_{ij} : force per area; symmetric (3D has 6 independent components).

Strain tensor ε_{ij} : dimensionless measure of deformation; symmetric.

Hooke’s law: $\sigma_{ij} = C_{ijkl} \varepsilon_{kl}$ (linear elasticity).

Isotropic case: 2 elastic constants suffice. Standard pairs: - Young’s modulus E + Poisson ν - Bulk modulus K + shear modulus G - Lamé constants $\lambda + \mu$

Wave propagation: - Longitudinal: $c_L = \sqrt{((K + 4G/3)/\rho)}$ - Transverse: $c_T = \sqrt{(G/\rho)}$

Applications: structural engineering, seismology (S-waves + P-waves), acoustics, materials science.

BST framework: continuum mechanics inherits from substrate-fluid framework (Vol 6 Lyra) + substrate-K-type Casimir structure underlying lattice dynamics.

Level 3 — Graduate physics student / theorem-level

Cauchy stress tensor:

For continuous medium, traction vector $t = \sigma \cdot n$ (Cauchy’s theorem) on surface with normal n . Stress tensor σ has 9 components in 3D (reduce to 6 by symmetry $\sigma_{ij} = \sigma_{ji}$).

Linear elasticity:

For small deformations, displacement field $u(x)$ gives strain $\varepsilon_{ij} = (1/2)(\partial_i u_j + \partial_j u_i)$. Hooke’s law:

$$\sigma_{ij} = C_{ijkl} \varepsilon_{kl}$$

Symmetry constraints on C : - $C_{ijkl} = C_{jikl}$ (stress symmetry) - $C_{ijkl} = C_{ijlk}$ (strain symmetry) - $C_{ijkl} = C_{klij}$ (thermodynamic / Maxwell relations) - 21 independent for triclinic - 3 independent for cubic crystals - 2 independent for isotropic

Elastic wave equation:

Newton's law for elastic medium:

$$\rho \ddot{u}_i = \partial_j \sigma_{ij} = C_{ijkl} \partial_j \partial_k u_l$$

For isotropic: gives separate longitudinal + transverse wave equations.

Substrate-natural framework:

Per Vol 6 Lyra substrate stat mech framework + Vol 9 Ch 3 lattice + Vol 9 Ch 7 phonons: - Elastic moduli emerge from lattice dynamics + interatomic potentials - Substrate-K-type Casimir structure provides energy-scale framework - Specific material elastic constants multi-week per honest scope

Per Cal #21 honest scope: EMPIRICAL PASS (continuum mechanics validated across engineering + geophysics + materials science) + MECHANISM PATH ARTICULATED via substrate-fluid + lattice dynamics framework; full D-tier multi-week.

Per Cal #99 META-theorem: continuum mechanics is a substrate-derivation consequence of substrate-fluid + substrate-lattice framework, NOT a new Strong-Uniqueness criterion.

What this chapter does NOT claim

- BST does NOT predict specific material elastic constants at v0.3.1 — multi-week per Vol 9 cross-link
- Honest weak-foundation scope per Vol 8 scaffold acknowledgment
- Standard elasticity (Landau-Lifshitz Vol 7) preserved at full classical precision

Bibliography

1. R. Hooke (1660): Hooke's law (founding of linear elasticity).
2. A. Cauchy (1822-1828): stress tensor + Cauchy's theorem.
3. G. Lamé (1852): Lamé constants + isotropic elasticity.
4. L. D. Landau + E. M. Lifshitz: *Theory of Elasticity* (Vol 7 of Course of Theoretical Physics).
5. Vol 6 (Stat Mech, Lyra): substrate fluid framework.
6. Vol 9 Ch 3 + Ch 7: lattice + phonons cross-links.

— Elie, Vol 8 Ch 9 v0.3.1, 2026-05-23 Saturday (Calibration #23 substance refill: 45 → ~125 lines + Cauchy stress + Lamé constants + elastic wave equation + 3-level pedagogy expanded)